

17 (a) Show that the equation $x^3 + 2x - 6 = 0$ has a solution between $x = 1$ and $x = 2$

(2)

(b) Show that the equation $x^3 + 2x - 6 = 0$ can be rearranged to give $x = \frac{6}{x^2 + 2}$

(1)

(c) Starting with $x_0 = 1.45$

use the iteration formula $x_{n+1} = \frac{6}{x_n^2 + 2}$ twice to find an estimate

for the solution of $x^3 + 2x - 6 = 0$

Give your answer correct to 4 decimal places.

(3)

(Total for Question 17 is 6 marks)